

Object Oriented Design Quality Metrics

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [jqassistant](#)
- [notebook walks through examples for integrating various packages with Neo4j](#)
- [OO Design Quality Metrics](#)
- [A Validation of Martin's Metric](#)
- [Neo4j Python Driver](#)

Incoming Dependencies

Incoming dependencies are also denoted as "Fan-in", "Afferent Coupling" or "in-degree". These are the ones that use the listed package.

If these packages get changed, the incoming dependencies might be affected by the change. The more incoming dependencies, the harder it gets to change the code without the need to adapt the dependent code ("rigid code"). Even worse, it might affect the behavior of the dependent code in an unwanted way ("fragile code").

Since Java Packages are organized hierarchically, incoming dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 1a

- Show the top 20 Java Packages with the most incoming dependencies
- Set the "incomingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incomingDependenciesIncludingSubpackages
0	axon-messaging-4.11.2	org.axonframework.messaging	messaging	10051	29462	
1	axon-messaging-4.11.2	org.axonframework.eventhandling	eventhandling	6120	26258	
2	axon-messaging-4.11.2	org.axonframework.commandhandling	commandhandling	1945	6341	
3	axon-messaging-4.11.2	org.axonframework.serialization	serialization	1771	7231	
4	axon-messaging-4.11.2	org.axonframework.messaging.annotation	annotation	1111	4088	
5	axon-messaging-4.11.2	org.axonframework.common	common	1062	2534	
6	axon-messaging-4.11.2	org.axonframework.tracing	tracing	635	2923	
7	axon-messaging-4.11.2	org.axonframework.queryhandling	queryhandling	436	1748	
8	axon-configuration-4.11.2	org.axonframework.config	config	425	3142	
9	axon-messaging-4.11.2	org.axonframework.common.transaction	transaction	368	1103	
10	axon-modelling-4.11.2	org.axonframework.modelling.command	command	276	947	
11	axon-modelling-4.11.2	org.axonframework.modelling.saga	saga	263	1142	
12	axon-messaging-4.11.2	org.axonframework.messaging.unitofwork	unitofwork	260	1100	
13	axon-messaging-4.11.2	org.axonframework.deadline	deadline	256	1252	
14	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore.jdbc	jdbc	235	1288	
15	axon-messaging-4.11.2	org.axonframework.messaging.deadletter	deadletter	206	1164	
16	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore	eventstore	205	816	
17	axon-messaging-4.11.2	org.axonframework.monitoring	monitoring	199	531	
18	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing	eventsourcing	148	580	
19	axon-server-connector-4.11.2	org.axonframework.axonserver.connector	connector	122	598	

Table 1b

- Show the top 20 Java Packages including their sub-packages with the most incoming dependencies
- Set the property "incomingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incomingDe
0	axon-messaging-4.11.2	org.axonframework.messaging	messaging	8918	28144	
1	axon-messaging-4.11.2	org.axonframework.eventhandling	eventhandling	3836	14367	
2	axon-messaging-4.11.2	org.axonframework.serialization	serialization	1550	6288	
3	axon-messaging-4.11.2	org.axonframework.common	common	1478	4132	
4	axon-messaging-4.11.2	org.axonframework.commandhandling	commandhandling	1058	4017	
5	axon-messaging-4.11.2	org.axonframework.messaging.annotation	annotation	994	3745	
6	axon-messaging-4.11.2	org.axonframework.tracing	tracing	565	2627	
7	axon-messaging-4.11.2	org.axonframework.common.transaction	transaction	362	1089	
8	axon-configuration-4.11.2	org.axonframework.config	config	302	1907	
9	axon-messaging-4.11.2	org.axonframework.queryhandling	queryhandling	273	1055	
10	axon-messaging-4.11.2	org.axonframework.messaging.unitofwork	unitofwork	223	883	
11	axon-messaging-4.11.2	org.axonframework.monitoring	monitoring	189	501	
12	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing	eventsourcing	173	645	
13	axon-messaging-4.11.2	org.axonframework.messaging.deadletter	deadletter	173	1030	
14	axon-modelling-4.11.2	org.axonframework.modelling.command	command	162	535	
15	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore	eventstore	144	464	
16	axon-messaging-4.11.2	org.axonframework.deadline	deadline	137	773	
17	axon-modelling-4.11.2	org.axonframework.modelling.saga	saga	97	277	
18	axon-messaging-4.11.2	org.axonframework.lifecycle	lifecycle	95	262	
19	axon-messaging-4.11.2	org.axonframework.messaging.responsetypes	responsetypes	90	372	

Outgoing Dependencies

Outgoing dependencies are also denoted as "Fan-out", "Efferent Coupling" or "out-degree". These are the ones that are used by the listed package.

Code from other packages and libraries you're depending on (outgoing) might change over time. The more outgoing changes, the more likely and frequently code changes are needed. This involves time and effort which can be reduced by automation of tests and version updates. Automated tests are crucial to reveal updates, that change the behavior of the

code unexpectedly (“fragile code”). As soon as more effort is required, keeping up becomes difficult (“rigid code”). Not being able to use a newer version might not only restrict features, it can get problematic if there are security issues. This might force you to take “fast but ugly” solutions into account which further increases technical dept.

Since Java Packages are organized hierarchically, outgoing dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 2a

- Show the top 20 Java Packages with the most outgoing dependencies
- Set the "outgoingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgoingDependenciesIncludingSubpackages
0	axon-configuration-4.11.2	org.axonframework.config	config	9977	38369	
1	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.autoconfig	autoconfig	6403	23156	
2	axon-test-4.11.2	org.axonframework.test.aggregate	aggregate	2210	7978	
3	axon-messaging-4.11.2	org.axonframework.eventhandling	eventhandling	1563	6457	
4	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.query	query	1544	7134	
5	axon-disruptor-4.11.2	org.axonframework.disruptor.commandhandling	commandhandling	1453	5180	
6	axon-test-4.11.2	org.axonframework.test.saga	saga	1419	4142	
7	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore.jdbc	jdbc	1340	6712	
8	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event.axon	axon	1178	4258	
9	axon-messaging-4.11.2	org.axonframework.queryhandling	queryhandling	1141	5254	
10	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing	eventsourcing	1009	3348	
11	axon-messaging-4.11.2	org.axonframework.eventhandling.pooled	pooled	962	4467	
12	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.command	command	932	3060	
13	axon-modelling-4.11.2	org.axonframework.modelling.command	command	847	3239	
14	axon-modelling-4.11.2	org.axonframework.modelling.command.inspection	inspection	793	3001	
15	axon-messaging-4.11.2	org.axonframework.commandhandling.distributed	distributed	738	2227	
16	axon-server-connector-4.11.2	org.axonframework.axonserver.connector	connector	721	3674	
17	axon-messaging-4.11.2	org.axonframework.commandhandling	commandhandling	691	1832	
18	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore	eventstore	613	1981	
19	axon-messaging-4.11.2	org.axonframework.eventhandling.deadletter.jdbc	jdbc	560	3518	

Table 2b

- Show the top 20 Java Packages including their sub-packages with the most outgoing dependencies
- Set the property "outgoingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgoingDependenciesCount
0	axon-configuration-4.11.2	org.axonframework.config	config	595	1832	1
1	axon-messaging-4.11.2	org.axonframework.eventhandling	eventhandling	588	2124	1
2	axon-server-connector-4.11.2	org.axonframework.axonserver.connector	connector	416	1523	1
3	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.autoconfig	autoconfig	401	1313	1
4	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing	eventsourcing	380	1319	1
5	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot	springboot	359	1156	1
6	axon-test-4.11.2	org.axonframework.test	test	273	793	1
7	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore	eventstore	245	914	1
8	axon-modelling-4.11.2	org.axonframework.modelling.command	command	232	754	1
9	axon-messaging-4.11.2	org.axonframework.eventhandling.deadletter	deadletter	207	996	1
10	axon-messaging-4.11.2	org.axonframework.commandhandling	commandhandling	196	485	1
11	axon-messaging-4.11.2	org.axonframework.messaging	messaging	193	499	1
12	axon-messaging-4.11.2	org.axonframework.deadline	deadline	190	611	1
13	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.query	query	179	707	1
14	axon-messaging-4.11.2	org.axonframework.queryhandling	queryhandling	177	614	1
15	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event	event	167	548	1
16	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event.axon	axon	167	555	1
17	axon-modelling-4.11.2	org.axonframework.modelling.saga	saga	155	474	1
18	axon-test-4.11.2	org.axonframework.test.aggregate	aggregate	149	548	1
19	axon-disruptor-4.11.2	org.axonframework.disruptor.commandhandling	commandhandling	130	436	1

Instability

$$Instability = \frac{Outgoing\ Dependencies}{Outgoing\ Dependencies + Incoming\ Dependencies}$$

Instability is expressed as the ratio of the number of outgoing dependencies of a module (i.e., the number of packages that depend on it) to the total number of dependencies (i.e., the sum of incoming and outgoing dependencies).

Small values near zero indicate low *Instability*. With no outgoing but some incoming dependencies the *Instability* is zero which is denoted as maximally stable. Such code units are more rigid and difficult to change without impacting other parts of the system. If they are changed less because of that, they are considered stable.

Conversely, high values approaching one indicate high *Instability*. With some outgoing dependencies but no incoming ones the *Instability* is denoted as maximally unstable. Such code units are easier to change without affecting other modules, making them more flexible and less prone to cascading changes throughout the system. If they are changed more often because of that, they are considered unstable.

Since Java Packages are organized hierarchically, *Instability* can be calculated for every package in isolation or by including all of its sub-packages.

Table 3a

- Show the top 20 Java Packages with the lowest *Instability*
- Set the property "instability" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityPe
0	axon-messaging-4.11.2	org.axonframework.messaging	messaging	0.013060	0.086849	0.180723	0
1	axon-messaging-4.11.2	org.axonframework.common.transaction	transaction	0.016043	0.043011	0.000000	0
2	axon-messaging-4.11.2	org.axonframework.common	common	0.021198	0.038860	0.000000	0
3	axon-messaging-4.11.2	org.axonframework.eventhandling.scheduling	scheduling	0.074074	0.111111	0.000000	0
4	axon-messaging-4.11.2	org.axonframework.lifecycle	lifecycle	0.090090	0.184211	0.000000	0
5	axon-messaging-4.11.2	org.axonframework.serialization	serialization	0.090395	0.201754	0.291667	0
6	axon-messaging-4.11.2	org.axonframework.monitoring	monitoring	0.095455	0.145833	0.333333	0
7	axon-messaging-4.11.2	org.axonframework.common.annotation	annotation	0.120000	0.120000	0.000000	0
8	axon-messaging-4.11.2	org.axonframework.common.stream	stream	0.131579	0.150000	0.000000	0
9	axon-messaging-4.11.2	org.axonframework.tracing	tracing	0.161162	0.198529	0.480000	0
10	axon-messaging-4.11.2	org.axonframework.eventhandling	eventhandling	0.203436	0.324544	0.495652	0
11	axon-messaging-4.11.2	org.axonframework.messaging.annotation	annotation	0.212057	0.292793	0.406250	0
12	axon-messaging-4.11.2	org.axonframework.common.jpa	jpa	0.240000	0.217391	1.000000	0
13	axon-messaging-4.11.2	org.axonframework.common.legacyjpa	legacyjpa	0.260870	0.238095	1.000000	0
14	axon-messaging-4.11.2	org.axonframework.commandhandling	commandhandling	0.262140	0.345794	0.600000	0
15	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot	springboot	0.294737	0.460000	0.000000	0
16	axon-messaging-4.11.2	org.axonframework.messaging.correlation	correlation	0.304348	0.200000	0.400000	0
17	axon-messaging-4.11.2	org.axonframework.serialization.upcasting	upcasting	0.312500	0.083333	0.000000	0
18	axon-messaging-4.11.2	org.axonframework.messaging.unitofwork	unitofwork	0.321149	0.194175	0.583333	0
19	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event.util	util	0.333333	0.250000	0.000000	0

Table 3b

- Show the top 20 Java Packages including their sub-packages with the lowest *Instability*
- Set the property "instabilityIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityPe
0	axon-messaging-4.11.2	org.axonframework.common	common	0.003372	0.012048	0.000000	0
1	axon-messaging-4.11.2	org.axonframework.messaging	messaging	0.021183	0.075610	0.402878	0
2	axon-messaging-4.11.2	org.axonframework.serialization	serialization	0.033063	0.154839	0.181818	0
3	axon-messaging-4.11.2	org.axonframework.lifecycle	lifecycle	0.040404	0.068966	0.000000	0
4	axon-messaging-4.11.2	org.axonframework.monitoring	monitoring	0.040609	0.052632	0.454545	0
5	axon-messaging-4.11.2	org.axonframework.tracing	tracing	0.055184	0.141414	0.187500	0
6	axon-messaging-4.11.2	org.axonframework.messaging.annotation	annotation	0.071028	0.151515	0.424242	0
7	axon-messaging-4.11.2	org.axonframework.common.annotation	annotation	0.086957	0.086957	0.000000	0
8	axon-messaging-4.11.2	org.axonframework.common.lock	lock	0.107143	0.142857	0.000000	0
9	axon-messaging-4.11.2	org.axonframework.messaging.unitofwork	unitofwork	0.111554	0.100000	0.375000	0
10	axon-messaging-4.11.2	org.axonframework.messaging.responsetypes	responsetypes	0.126214	0.333333	0.000000	0
11	axon-messaging-4.11.2	org.axonframework.eventhandling	eventhandling	0.132911	0.375887	0.452174	0
12	axon-messaging-4.11.2	org.axonframework.common.jdbc	jdbc	0.150943	0.212121	0.000000	0
13	axon-messaging-4.11.2	org.axonframework.common.property	property	0.153846	0.222222	0.000000	0
14	axon-messaging-4.11.2	org.axonframework.commandhandling	commandhandling	0.156300	0.478571	0.512195	0
15	axon-messaging-4.11.2	org.axonframework.common.jpa	jpa	0.157895	0.117647	0.000000	0
16	axon-messaging-4.11.2	org.axonframework.common.legacyjpa	legacyjpa	0.176471	0.133333	0.000000	0
17	axon-messaging-4.11.2	org.axonframework.messaging.deadletter	deadletter	0.183962	0.272727	0.466667	0
18	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.snapshotting	snapshotting	0.187500	0.250000	0.500000	0
19	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event.util	util	0.200000	0.200000	0.000000	0

Abstractness

$$Abstractness = \frac{\text{abstract classes in category}}{\text{total number of classes in category}}$$

Package *Abstractness* is expressed as the ratio of the number of abstract classes and interfaces to the total number of classes of a package.

Zero *Abstractness* means that there are no abstract types or interfaces in the package. On the other hand, a value of one means that there are only abstract types.

Since Java Packages are organized hierarchically, *Abstractness* can be calculated for every package in isolation or by including all of its sub-packages.

Table 4a

- Show the top 30 packages with the lowest *Abstractness*
- Set the property "abstractness" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
0	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.autoconfigure	autoconfig	0.0	0	39
1	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot	springboot	0.0	0	28
2	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore.legacyjpa	legacyjpa	0.0	0	10
3	axon-messaging-4.11.2	org.axonframework.commandhandling.distributed.commandfilter	commandfilter	0.0	0	7
4	axon-messaging-4.11.2	org.axonframework.deadline.dbscheduler	dbscheduler	0.0	0	7
5	axon-messaging-4.11.2	org.axonframework.eventhandling.scheduling.dbscheduler	dbscheduler	0.0	0	7
6	axon-messaging-4.11.2	org.axonframework.messaging.timeout	timeout	0.0	0	7
7	axon-messaging-4.11.2	org.axonframework.serialization.json	json	0.0	0	7
8	axon-messaging-4.11.2	org.axonframework.serialization.xml	xml	0.0	0	7
9	axon-messaging-4.11.2	org.axonframework.tracing.attributes	attributes	0.0	0	6
10	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.query.subscription	subscription	0.0	0	6
11	axon-messaging-4.11.2	org.axonframework.eventhandling.deadletter	deadletter	0.0	0	5
12	axon-messaging-4.11.2	org.axonframework.serialization.converters	converters	0.0	0	5
13	axon-tracing-opentelemetry-4.11.2	org.axonframework.tracing.opentelemetry	opentelemetry	0.0	0	5
14	axon-messaging-4.11.2	org.axonframework.commandhandling.callbacks	callbacks	0.0	0	4
15	axon-messaging-4.11.2	org.axonframework.deadline.quartz	quartz	0.0	0	4
16	axon-messaging-4.11.2	org.axonframework.eventhandling.scheduling.java	java	0.0	0	4
17	axon-messaging-4.11.2	org.axonframework.eventhandling.tokenstore.jpa	jpa	0.0	0	4
18	axon-test-4.11.2	org.axonframework.test.server	server	0.0	0	4
19	axon-modelling-4.11.2	org.axonframework.modelling.saga.repository.legacyjpa	legacyjpa	0.0	0	3
20	axon-messaging-4.11.2	org.axonframework.eventhandling.scheduling.jobrunr	jobrunr	0.0	0	3
21	axon-messaging-4.11.2	org.axonframework.util	util	0.0	0	3
22	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.processor	processor	0.0	0	3
23	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event.util	util	0.0	0	3
24	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.actuator.axonserver	axonserver	0.0	0	2
25	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.autoconfigure.legacyjpa	legacyjpa	0.0	0	2
26	axon-modelling-4.11.2	org.axonframework.modelling.command.legacyjpa	legacyjpa	0.0	0	2
27	axon-modelling-4.11.2	org.axonframework.modelling.saga.repository.inmemory	inmemory	0.0	0	2
28	axon-messaging-4.11.2	org.axonframework.eventhandling.tokenstore.inmemory	inmemory	0.0	0	2
29	axon-messaging-4.11.2	org.axonframework.eventhandling.tokenstore.legacyjpa	legacyjpa	0.0	0	2

Table 4b

- Show the top 30 packages with the highest *Abstractness* and number of Java Types

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
139	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore.jdbc...	statements	1.000000	15	15
140	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.actuator	actuator	1.000000	1	1
138	axon-messaging-4.11.2	org.axonframework.serialization.upcasting	upcasting	0.833333	5	6
135	axon-messaging-4.11.2	org.axonframework.common.annotation	annotation	0.666667	2	3
136	axon-messaging-4.11.2	org.axonframework.common.stream	stream	0.666667	2	3
137	axon-messaging-4.11.2	org.axonframework.eventhandling.scheduling	scheduling	0.666667	2	3
134	axon-messaging-4.11.2	org.axonframework.common	common	0.642857	18	28
131	axon-messaging-4.11.2	org.axonframework.messaging	messaging	0.600000	21	35
132	axon-messaging-4.11.2	org.axonframework.lifecycle	lifecycle	0.600000	6	10
133	axon-messaging-4.11.2	org.axonframework.eventhandling.gateway	gateway	0.600000	3	5
130	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.util	util	0.555556	5	9
127	axon-messaging-4.11.2	org.axonframework.serialization.upcasting.event	event	0.500000	6	12
128	axon-modelling-4.11.2	org.axonframework.modelling.saga.metamodel	metamodel	0.500000	2	4
129	axon-messaging-4.11.2	org.axonframework.common.transaction	transaction	0.500000	2	4
126	axon-configuration-4.11.2	org.axonframework.config	config	0.452381	19	42
124	axon-messaging-4.11.2	org.axonframework.common.jdbc	jdbc	0.444444	8	18
125	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.conflictresolu...	conflictresolution	0.444444	4	9
123	axon-modelling-4.11.2	org.axonframework.modelling.command	command	0.410714	23	56
122	axon-messaging-4.11.2	org.axonframework.eventhandling	eventhandling	0.410000	41	100
117	axon-messaging-4.11.2	org.axonframework.deadline.annotation	annotation	0.400000	2	5
118	axon-messaging-4.11.2	org.axonframework.queryhandling.registration	registration	0.400000	2	5
119	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.heartbeat	heartbeat	0.400000	2	5
120	axon-test-4.11.2	org.axonframework.test.eventscheduler	eventscheduler	0.400000	2	5
121	axon-test-4.11.2	org.axonframework.test.utils	utils	0.400000	2	5
116	axon-modelling-4.11.2	org.axonframework.modelling.saga	saga	0.393939	13	33
113	axon-messaging-4.11.2	org.axonframework.commandhandling	commandhandling	0.375000	12	32
114	axon-messaging-4.11.2	org.axonframework.messaging.interceptors	interceptors	0.375000	3	8
115	axon-messaging-4.11.2	org.axonframework.messaging.responsetypes	responsetypes	0.375000	3	8
112	axon-messaging-4.11.2	org.axonframework.messaging.deadletter	deadletter	0.368421	7	19
111	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore	eventstore	0.354839	11	31

Table 4c

- Show the top 30 packages including their sub-packages with the highest package depth and lowest *Abstractness*

- Set the property "abstractnessIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSubpa
0	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.autoconfig	autoconfig	0.0	0	41	
1	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.heartbe...	connection	0.0	0	2	
2	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore.leg...	legacyjpa	0.0	0	10	
3	axon-messaging-4.11.2	org.axonframework.commandhandling.distributed....	commandfilter	0.0	0	7	
4	axon-messaging-4.11.2	org.axonframework.deadline.dbscheduler	dbscheduler	0.0	0	7	
5	axon-messaging-4.11.2	org.axonframework.eventhandling.scheduling.dbs...	dbscheduler	0.0	0	7	
6	axon-messaging-4.11.2	org.axonframework.messaging.timeout	timeout	0.0	0	7	
7	axon-messaging-4.11.2	org.axonframework.serialization.json	json	0.0	0	7	
8	axon-messaging-4.11.2	org.axonframework.serialization.xml	xml	0.0	0	7	
9	axon-messaging-4.11.2	org.axonframework.tracing.attributes	attributes	0.0	0	6	
10	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.query.s...	subscription	0.0	0	6	
11	axon-messaging-4.11.2	org.axonframework.serialization.converters	converters	0.0	0	5	
12	axon-tracing-opentelemetry-4.11.2	org.axonframework.tracing.opentelemetry	opentelemetry	0.0	0	5	
13	axon-messaging-4.11.2	org.axonframework.commandhandling.callbacks	callbacks	0.0	0	4	
14	axon-messaging-4.11.2	org.axonframework.deadline.quartz	quartz	0.0	0	4	
15	axon-messaging-4.11.2	org.axonframework.eventhandling.scheduling.java	java	0.0	0	4	
16	axon-messaging-4.11.2	org.axonframework.eventhandling.tokenstore.jpa	jpa	0.0	0	4	
17	axon-test-4.11.2	org.axonframework.test.server	server	0.0	0	4	
18	axon-modelling-4.11.2	org.axonframework.modelling.saga.repository.le...	legacyjpa	0.0	0	3	
19	axon-messaging-4.11.2	org.axonframework.eventhandling.scheduling.job...	jobrunr	0.0	0	3	
20	axon-messaging-4.11.2	org.axonframework.util	util	0.0	0	3	
21	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.processor	processor	0.0	0	3	
22	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event.util	util	0.0	0	3	
23	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.actuator.axonserver	axonserver	0.0	0	2	
24	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.autoconfig.legacyjpa	legacyjpa	0.0	0	2	
25	axon-modelling-4.11.2	org.axonframework.modelling.command.legacyjpa	legacyjpa	0.0	0	2	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSubpa
26	axon-modelling-4.11.2	org.axonframework.modelling.saga.repository.in...	inmemory	0.0	0	2	
27	axon-messaging-4.11.2	org.axonframework.eventhandling.tokenstore.inm...	inmemory	0.0	0	2	
28	axon-messaging-4.11.2	org.axonframework.eventhandling.tokenstore.leg...	legacyjpa	0.0	0	2	
29	axon-messaging-4.11.2	org.axonframework.messaging.interceptors.legac...	legacyvalidation	0.0	0	2	

Table 4d

- Show the top 30 packages including their sub-packages with the highest package depth and highest *Abstractness*

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSubpa
118	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore.jdbc...	statements	1.000000	15	15	
116	axon-messaging-4.11.2	org.axonframework.common.annotation	annotation	0.666667	2	3	
117	axon-messaging-4.11.2	org.axonframework.common.stream	stream	0.666667	2	3	
115	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore.jdbc	jdbc	0.620690	18	29	
114	axon-messaging-4.11.2	org.axonframework.serialization.upcasting	upcasting	0.611111	11	18	
112	axon-messaging-4.11.2	org.axonframework.lifecycle	lifecycle	0.600000	6	10	
113	axon-messaging-4.11.2	org.axonframework.eventhandling.gateway	gateway	0.600000	3	5	
109	axon-messaging-4.11.2	org.axonframework.serialization.upcasting.event	event	0.500000	6	12	
110	axon-modelling-4.11.2	org.axonframework.modelling.saga.metamodel	metamodel	0.500000	2	4	
111	axon-messaging-4.11.2	org.axonframework.common.transaction	transaction	0.500000	2	4	
108	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.util	util	0.454545	5	11	
107	axon-configuration-4.11.2	org.axonframework.config	config	0.452381	19	42	
105	axon-messaging-4.11.2	org.axonframework.common.jdbc	jdbc	0.444444	8	18	
106	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.conflictresolu...	conflictresolution	0.444444	4	9	
104	axon-messaging-4.11.2	org.axonframework.common	common	0.425743	43	101	
100	axon-messaging-4.11.2	org.axonframework.deadline.annotation	annotation	0.400000	2	5	
101	axon-messaging-4.11.2	org.axonframework.queryhandling.registration	registration	0.400000	2	5	
102	axon-test-4.11.2	org.axonframework.test.eventscheduler	eventscheduler	0.400000	2	5	
103	axon-test-4.11.2	org.axonframework.test.utils	utils	0.400000	2	5	
99	axon-modelling-4.11.2	org.axonframework.modelling.command	command	0.380952	32	84	
98	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore	eventstore	0.379747	30	79	
97	axon-messaging-4.11.2	org.axonframework.messaging.responsetypes	responsetypes	0.375000	3	8	
96	axon-messaging-4.11.2	org.axonframework.messaging.deadletter	deadletter	0.368421	7	19	
95	axon-messaging-4.11.2	org.axonframework.messaging	messaging	0.364238	55	151	
94	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing	eventsourcing	0.353383	47	133	
93	axon-modelling-4.11.2	org.axonframework.modelling.command.inspection	inspection	0.346154	9	26	
88	axon-spring-boot-	org.axonframework.actuator	actuator	0.333333	1	3	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSubpa
	autoconfigure-4.11.2						
89	axon-test-4.11.2	org.axonframework.test.saga	saga	0.333333	7	21	
90	axon-messaging-4.11.2	org.axonframework.common.property	property	0.333333	3	9	
91	axon-messaging-4.11.2	org.axonframework.monitoring	monitoring	0.333333	2	6	

Distance from the main sequence

The *main sequence* is a imaginary line that represents a good compromise between *Abstractness* and *Instability*. A high distance to this line may indicate problems. For example is very *stable* (rigid) code with low abstractness hard to change.

Read more details on that in [OO Design Quality Metrics](#) and [Calculate metrics](#).

Table 5a

- Show the top 30 packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-messaging-4.11.2	org.axonframework.serialization.converters	converters	NaN	0.0	NaN	5
1	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.actuator	actuator	NaN	1.0	NaN	1
2	axon-messaging-4.11.2	org.axonframework.common.io	io	NaN	0.0	NaN	1
3	axon-messaging-4.11.2	org.axonframework.eventhandling.interceptors	interceptors	NaN	0.0	NaN	1
4	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.heartbe...	source	NaN	0.0	NaN	1
5	axon-spring-boot-autoconfigure-4.11.2	org	org	NaN	0.0	NaN	0
6	axon-spring-boot-autoconfigure-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0
7	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.service	service	NaN	0.0	NaN	0
8	axon-modelling-4.11.2	org	org	NaN	0.0	NaN	0
9	axon-modelling-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0
10	axon-modelling-4.11.2	org.axonframework.modelling	modelling	NaN	0.0	NaN	0
11	axon-configuration-4.11.2	org	org	NaN	0.0	NaN	0
12	axon-configuration-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0
13	axon-messaging-4.11.2	org	org	NaN	0.0	NaN	0
14	axon-messaging-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0
15	axon-server-connector-4.11.2	org	org	NaN	0.0	NaN	0
16	axon-server-connector-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0
17	axon-server-connector-4.11.2	org.axonframework.axonserver	axonserver	NaN	0.0	NaN	0
18	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.heartbe...	connection	NaN	0.0	NaN	0
19	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event	event	NaN	0.0	NaN	0
20	axon-tracing-opentelemetry-4.11.2	org	org	NaN	0.0	NaN	0
21	axon-tracing-opentelemetry-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0
22	axon-tracing-opentelemetry-4.11.2	org.axonframework.tracing	tracing	NaN	0.0	NaN	0
23	axon-disruptor-4.11.2	org	org	NaN	0.0	NaN	0
24	axon-disruptor-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0
25	axon-disruptor-4.11.2	org.axonframework.disruptor	disruptor	NaN	0.0	NaN	0
26	axon-eventsourcing-4.11.2	org	org	NaN	0.0	NaN	0
27	axon-eventsourcing-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0
28	axon-test-4.11.2	org	org	NaN	0.0	NaN	0
29	axon-test-4.11.2	org.axonframework	axonframework	NaN	0.0	NaN	0

Table 5b

- Show the top 30 packages including their sub-packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-server-connector-4.11.2	org.axonframework.axonserver.connector.event.util	util	0.800000	0.000000	0.200000	3
1	axon-messaging-4.11.2	org.axonframework.tracing	tracing	0.744816	0.200000	0.055184	25
2	axon-messaging-4.11.2	org.axonframework.common.lock	lock	0.711039	0.181818	0.107143	11
3	axon-messaging-4.11.2	org.axonframework.serialization	serialization	0.686449	0.280488	0.033063	82
4	axon-test-4.11.2	org.axonframework.test.server	server	0.666667	0.000000	0.333333	4
5	axon-messaging-4.11.2	org.axonframework.common.digest	digest	0.666667	0.000000	0.333333	1
6	axon-messaging-4.11.2	org.axonframework.messaging.annotation	annotation	0.632676	0.296296	0.071028	54
7	axon-messaging-4.11.2	org.axonframework.monitoring	monitoring	0.626058	0.333333	0.040609	6
8	axon-messaging-4.11.2	org.axonframework.messaging	messaging	0.614578	0.364238	0.021183	151
9	axon-test-4.11.2	org.axonframework.test.matchers	matchers	0.606707	0.125000	0.268293	24
10	axon-messaging-4.11.2	org.axonframework.messaging.unitofwork	unitofwork	0.602732	0.285714	0.111554	14
11	axon-messaging-4.11.2	org.axonframework.eventhandling	eventhandling	0.592827	0.274262	0.132911	237
12	axon-messaging-4.11.2	org.axonframework.common.jpa	jpa	0.592105	0.250000	0.157895	4
13	axon-messaging-4.11.2	org.axonframework.common.legacyjpa	legacyjpa	0.573529	0.250000	0.176471	4
14	axon-messaging-4.11.2	org.axonframework.common	common	0.570886	0.425743	0.003372	101
15	axon-messaging-4.11.2	org.axonframework.commandhandling	commandhandling	0.562147	0.281553	0.156300	103
16	axon-messaging-4.11.2	org.axonframework.messaging.correlation	correlation	0.533784	0.250000	0.216216	4
17	axon-messaging-4.11.2	org.axonframework.common.property	property	0.512821	0.333333	0.153846	9
18	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.util.legacyjpa	legacyjpa	0.500000	0.000000	0.500000	1
19	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.springboot.util.jpa	jpa	0.500000	0.000000	0.500000	1
20	axon-messaging-4.11.2	org.axonframework.messaging.responsetypes	responsetypes	0.498786	0.375000	0.126214	8
21	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.eventstore.jdbc...	statements	0.480769	1.000000	0.480769	15
22	axon-eventsourcing-4.11.2	org.axonframework.eventsourcing.snapshotting	snapshotting	0.479167	0.333333	0.187500	3
23	axon-messaging-4.11.2	org.axonframework.serialization.xml	xml	0.466667	0.000000	0.533333	7
24	axon-messaging-4.11.2	org.axonframework.util	util	0.454545	0.000000	0.545455	3
25	axon-messaging-4.11.2	org.axonframework.messaging.deadletter	deadletter	0.447617	0.368421	0.183962	19
26	axon-messaging-4.11.2	org.axonframework.commandhandling.callbacks	callbacks	0.409091	0.000000	0.590909	4
27	axon-messaging-4.11.2	org.axonframework.common.jdbc	jdbc	0.404612	0.444444	0.150943	18
28	axon-spring-boot-autoconfigure-4.11.2	org.axonframework.actuator.axonserver	axonserver	0.400000	0.000000	0.600000	2
29	axon-modelling-4.11.2	org.axonframework.modelling.saga.repository.in...	inmemory	0.400000	0.000000	0.600000	2

Abstractness vs. Instability Plot with "Main Sequence" line as reference

- Plot *Abstractness* vs. *Instability* of all packages
- Draw the "main sequence" as dashed green diagonal line
- Scale the packages by the number of types they contain

- Color the packages by their distance to the "main sequence" (blue=near, red=far)

Figure 5a - Packages without their sub-packages

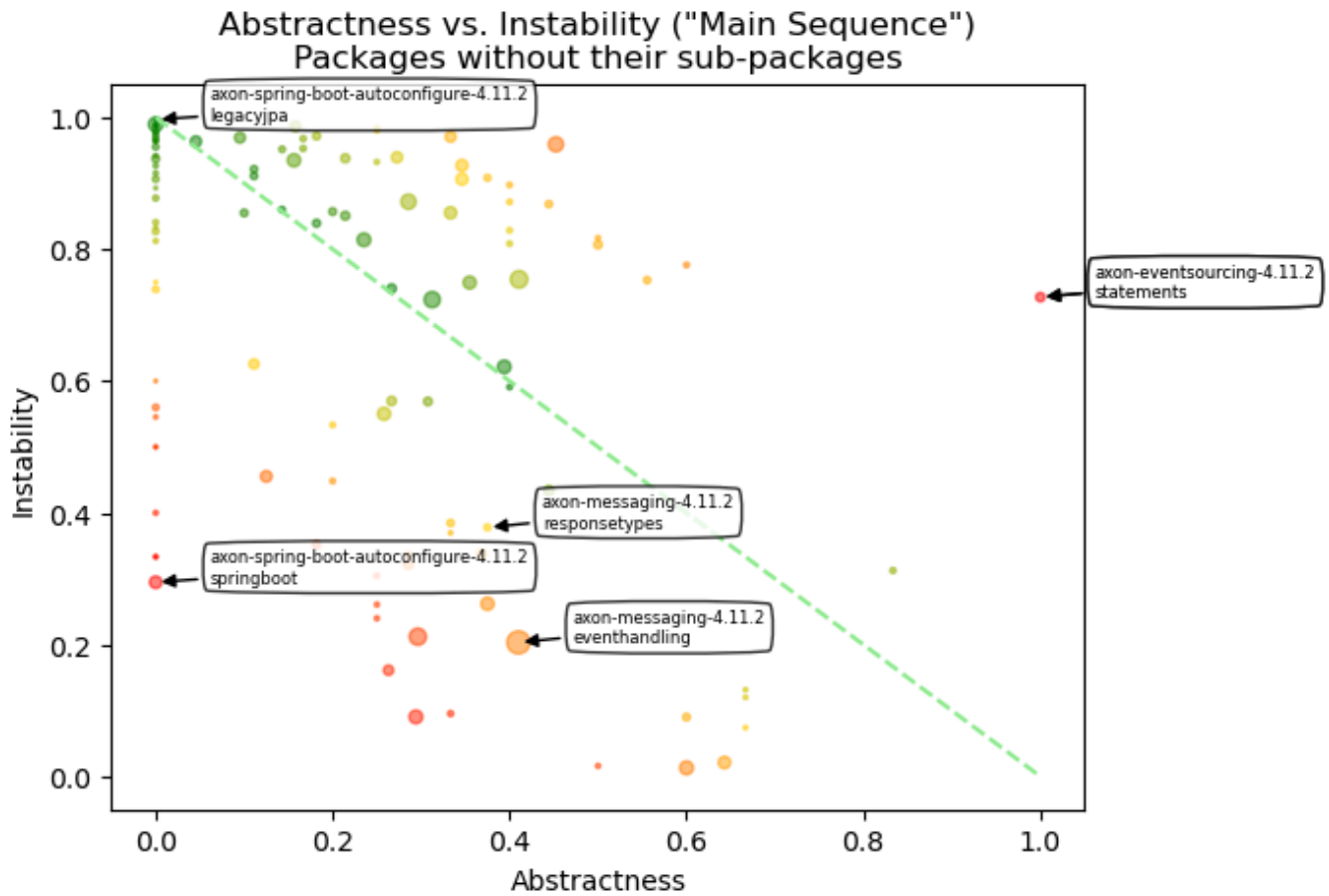


Figure 5b - Packages including their sub-packages

Abstractness vs. Instability ("Main Sequence")
Packages including their sub-packages

